EE1-10A MATHEMATICS I

1. (a) Express in the form $x+iy$: [Seen similar.]

$$(i)\quad \frac{\sin\frac{\pi}{3}+i\cos\frac{\pi}{3}}{\cos\frac{\pi}{6}+i\sin\frac{\pi}{6}}, \qquad (ii)\quad \left(\frac{i+1}{i-1}\right)^{2019} \qquad (iii)\quad (-i)^{-i}$$

SOLUTION

Each time, express in complex exponential form:

(i) $$\frac{\sin\frac{\pi}{3}+i\cos\frac{\pi}{3}}{\cos\frac{\pi}{6}+i\sin\frac{\pi}{6}} = \frac{i(\cos\frac{\pi}{3}-i\sin\frac{\pi}{3})}{\cos\frac{\pi}{6}+i\sin\frac{\pi}{6}} = i\left(\frac{e^{-i\pi/3)}}{e^{i\pi/6}}\right) = ie^{-i\pi/2} = i(-i) = 1$$

(ii) $$\left(\frac{i+1}{i-1}\right)^{2019} = \left(\frac{\sqrt{2}e^{i\pi/4}}{\sqrt{2}e^{i3\pi/4}}\right)^{2019} = \left(e^{-i\pi/2}\right)^{2019} = (-i)^{2019} = (-i)^3(-i)^{2016} = -i$$

(iii) $$(-i)^{-i} = \left(e^{i(-\pi/2+2n\pi)}\right)^{-i} = e^{-i^2(-\pi/2+2n\pi)} = e^{-\pi/2+2n\pi}$$, for integer n.

(b) (i) Use the definitions of sinh and cosh in terms of exponentials, to prove that [Bookwork]

$$\sinh(x+iy) = \sinh x\cos y + i\cosh x\sin y\,.$$

SOLUTION Beginning by expressing as exponentials:

$$\sinh(x+iy) = \frac{e^{x+iy}-e^{-(x+iy)}}{2} = \frac{e^xe^{iy}-e^{-x}e^{-iy}}{2} = \frac{e^x(\cos y+i\sin y)-e^{-x}(\cos y-i\sin y)}{2}$$

$$= \frac{1}{2}\left(e^x-e^{-x}\right)\cos y + i\frac{1}{2}\left(e^x+e^{-x}\right)\sin y = \sinh x\cos y + i\cosh x\sin y\,,$$

as required.

(ii) Hence, or otherwise, find all complex roots of [Seen similar]

$$\sinh^2(z) = -1\,.$$

SOLUTION Begin with $\sinh^2(z) = -1 \Rightarrow \sinh(z) = \sinh(x+iy) = \pm i$, and use (i) to rewrite as

$$\sinh x\cos y + i\cosh x\sin y = \pm i\,.$$

Equating real and imaginary parts:

$$\sinh x\cos y = 0 \quad (1), \quad \text{and} \quad \cosh x\sin y = \pm 1 \quad (2).$$

Begin with (1):

$$\sinh x\cos y = 0 \Rightarrow \sinh x = 0 \quad \text{or} \quad \cos y = 0$$

so either $x = 0$ or $y = (2n+1)\pi/2$ for integer n. Substitute these into (2):

$$x = 0: \quad \cosh 0\sin y = \pm 1 \Rightarrow \sin y = \pm 1 \Rightarrow y = (2n+1)\pi/2\,,$$

so no new solutions here. Similarly,

$$y = (2n+1)\pi/2: \quad \cosh x\sin\left(\frac{(2n+1)\pi}{2}\right) = \pm 1 \Rightarrow \cosh x = \pm 1\,.$$

As $\cosh x = -1$, has no solution and $\cosh x = 1 \Rightarrow x = 0$, we find no new solutions again, leaving $z = (2n+1)\pi i/2$.

(c) Find the following limits

(i) $\lim_{x\to 3} \dfrac{\sqrt{x-2}-1}{\sqrt{x^2-5}-2}$, [Seen similar]

SOLUTION

Can use l'Hopital, as of form "0/0":

$$\lim_{x\to 3} \frac{\sqrt{x-2}-1}{\sqrt{x^2-5}-2} = \lim_{x\to 3} \frac{\frac{1}{2\sqrt{x-2}}}{\frac{1}{2\sqrt{x^2-5}}2x} = \lim_{x\to 3} \frac{\sqrt{x^2-5}}{2x\sqrt{x-2}} = \frac{\sqrt{4}}{6\sqrt{1}} = \frac{1}{3}\,.$$

(ii) $\lim_{x\to\infty}\left(\frac{x+1}{x}\right)^x$, [New]

SOLUTION

Let $y=\left(\frac{x+1}{x}\right)^x$, then $\ln y = x\ln\left(1+\frac{1}{x}\right)$. As $x\to\infty$, clearly $\left|\frac{1}{x}\right|<1$ so can use series:

$$\ln y = x\left(\frac{1}{x}-\frac{1}{2x^2}+\frac{1}{3x^3}-\cdots\right). = 1-\frac{1}{2x}+\frac{1}{3x^2}-\cdots$$

Hence

$$\lim_{x\to\infty}\ln y = 1 \Rightarrow \lim_{x\to\infty} y = e\,.$$

(iii) $\lim_{x\to 0}\frac{\ln(1-2x^2)}{e^x-x-1}$ [Seen similar]

SOLUTION

Again, given that as $x\to 0$ we have $|-2x^2|<1$ we can apply Maclaurin series in numerator, as well as in denominator:

$$\lim_{x\to 0}\frac{\ln(1-2x^2)}{e^x-x-1} = \lim_{x\to 0}\frac{(-2x^2)-\frac{1}{2}(-2x^2)^2+\frac{1}{3}(-2x^2)^3+\cdots}{1+x+\frac{1}{2}x^2+\frac{1}{6}x^3+\frac{1}{4!}x^4+\cdots-x-1} = \lim_{x\to 0}\frac{-2x^2-2x^4+\frac{8}{3}x^6+\cdots}{\frac{1}{2}x^2+\frac{1}{6}x^3+\frac{1}{4!}x^4+\cdots}$$

$$= \lim_{x\to 0}\frac{-2-2x^2+\frac{8}{3}x^4+\cdots}{\frac{1}{2}+\frac{1}{6}x+\frac{1}{4!}x^2+\cdots} = -4\,.$$

(d)

A function is defined as

$$h(x)=\begin{cases} x^{-3}\tan^3(2x) & \text{for} \quad x\neq 0\,; \\ m & \text{for} \quad x=0\,, \end{cases}$$

What value of m makes $h(x)$ a continuous function? [Seen similar]

SOLUTION

For continuity, we need $\lim_{x\to 0}\frac{\tan^3(2x)}{x^3} = h(0) = m$, so we rewrite as

$\lim_{x\to 0}\left(\frac{\sin(2x)}{x}\right)\frac{1}{\cos^3(2)} = 2^3/1^3 = 8$, where we have assumed $\lim_{x\to 0}\sin(ax)/x = a$. Hence for continuity at $x=0$, we require $m=8$.

2 (a) Differentiate to obtain $\dfrac{dy}{dx}$: [Seen similar]

$$(i) \quad y = \frac{\ln x}{xe^x}, \qquad (ii) \quad y = \ln(x + \sqrt{x^2-1}), \qquad (iii) \quad x^2 + y^2 + e^{xy} = 1.$$

SOLUTION

(i) Quotient rule: $y' = \dfrac{f'g - fg'}{g^2} = \dfrac{\frac{1}{x}(xe^x) - \ln x(e^x + xe^x)}{(xe^x)^2} = \dfrac{1-(1+x)\ln x}{x^2 e^x}$.

(ii) Chain rule:

$$y' = \frac{1}{x+\sqrt{x^2-1}}\left(1 + \frac{2x}{2\sqrt{x^2-1}}\right) = \frac{1}{x+\sqrt{x^2-1}}\left(\frac{\sqrt{x^2-1}}{\sqrt{x^2-1}} + \frac{x}{\sqrt{x^2-1}}\right) = \frac{1}{\sqrt{x^2-1}}.$$

(iii) Implicit differentiation:

$$2x + 2y\frac{dy}{dx} + e^{xy}\left(y + x\frac{dy}{dx}\right) = 0 \Rightarrow (2y + xe^{xy})\frac{dy}{dx} = -(2x + ye^{xy}) \Rightarrow \frac{dy}{dx} = -\frac{2x+ye^{xy}}{2y+xe^{xy}}.$$

(b) Using the definition of the derivative as a limit (differentiation from first principles), differentiate $y = \dfrac{1}{x}$. [Bookwork]

SOLUTION

Differentiating from first principles:

$$\frac{dy}{dx} = \lim_{h\to 0}\frac{f(x+h)-f(x)}{h} = \lim_{h\to 0}\frac{\frac{1}{x+h} - \frac{1}{x}}{h} = \lim_{h\to 0}\frac{x-(x+h)}{x(x+h)h} = \lim_{h\to 0}\frac{-1}{x(x+h)} = -\frac{1}{x^2}.$$

(c) Given the function $y = e^{x^2/3}$, [Seen similar/New]

(i) Show that $\dfrac{dy}{dx} = \dfrac{2}{3}xy$.

SOLUTION Using chain rule:

$$\frac{dy}{dx} = e^{x^2/3}\left(\frac{2x}{3}\right) = \frac{2}{3}xy.$$

as required.

(ii) Differentiate the last equation n times, using Leibnitz' formula, and obtain a recurrence relation giving $y^{(n+1)}(0)$ in terms of $y^{(n-1)}(0)$.

[Note: $y^{(k)}(a)$ means y differentiated k times, and evaluated at $x = a$.]

SOLUTION

Use Leibnitz' theorem to differentiate n times:

$$D^n(y') = \frac{2}{3}D^n(xy) \Rightarrow y^{(n+1)}(x) = \frac{2}{3}\left(xy^{(n)}(x) + n(1)y^{(n-1)}(x) + 0 + 0 + \ldots\right)$$

Now substitute $x = 0$ to obtain the required recurrence relation:

$$y^{(n+1)}(0) = \frac{2}{3}ny^{(n-1)}(0).$$

(iii) Hence, or otherwise, evaluate $y^{(5)}(0)$.

SOLUTION To find the fifth derivative $y^{(5)}(0)$, we first need

$$y^{(0)}(0) = y(0) = e^0 = 1 \quad \text{and} \quad y^{(1)}(0) = y'(0) = \frac{2}{3}(0)y = 0.$$

Now use the recurrence relation, first with $n = 4$, then $n = 2$:

$$y^{(5)}(0) = \frac{2}{3}(4)y^{(3)}(0) = \frac{8}{3}\frac{2}{3}(2)y^{(1)}(0) = \frac{32}{9}0 = 0.$$

(d) Prove by induction that for $n \geq 0$,

$$D^n\left[\sin(ax)\right] = a^n \sin\left(ax + \frac{n\pi}{2}\right),$$

where D is the differential operator, $D = \dfrac{d}{dx}$. [Bookwork]

SOLUTION

Let the statement be $P(n)$. Begin with the base case, $n = 0$. The statement is

$$D^0[\sin(ax)] = a^0 \sin\left(ax + \frac{0\pi}{2}\right).$$

the lhs is $\sin(ax)$ differentiated zero times, so $\sin(ax)$, equal to the rhs, establishing the base case, $P(0)$.

For the inductive step, assume the statement is true for $n = k$, the statement

$$P(k): \quad D^k\left[\sin(ax)\right] = a^k \sin\left(ax + \frac{k\pi}{2}\right).$$

Then differentiate both sides, so

$$D^{k+1}\left[\sin(ax)\right] = \frac{d}{dx} D^k\left[\sin(ax)\right] = a^k \frac{d}{dx}\left[\sin\left(ax + \frac{k\pi}{2}\right)\right] = a^k \cdot a\cos\left(ax + \frac{k\pi}{2}\right)$$

and as $\cos\theta = \sin(\theta + \pi/2)$, in general, we have

$$D^{k+1}\left[\sin(ax)\right] = a^{k+1}\sin\left(ax + \frac{k\pi}{2} + \frac{\pi}{2}\right) = a^{k+1}\sin\left(ax + \frac{(k+1)\pi}{2}\right)$$

This is the statement for $n = k+1$. Hence, we have shown that $P(k) \Rightarrow P(k+1)$ and, together with the base case $P(0)$, the result is true for all integers $n \geq 0$, by mathematical induction.

3. (a) Evaluate the following integrals: [Seen similar]

(i) $\displaystyle\int \frac{\sin x-1}{\sin x+1}\,dx,$

SOLUTION

Multiply numerator and denominator by $\sin x-1$:

$$\int \frac{(\sin x-1)(\sin x-1)}{(\sin x+1)(\sin x-1)}\,dx = \int \frac{(\sin^2 x-2\sin x+1}{-\cos^2 x}\,dx = \int -(\tan^x+\sec^2 x)+\frac{2\sin x}{\cos^2 x}\,dx$$

$$\int -(2\sec^x-1)+\frac{2\sin x}{\cos^2 x}\,dx = -2\tan x+x+\frac{2}{\cos x}+C$$

where we have used the identity $\tan^2 x = \sec^2 x-1$. Alternatively, could use substitution $t = \tan(x/2)$.

(ii) $\displaystyle\int \frac{x^2+2}{x^3+x^2-2x}\,dx,$

SOLUTION

Partial fractions first:

$$\frac{x^2+2}{x(x^2+x-2)} = \frac{A}{x}+\frac{B}{x+2}+\frac{C}{x-1} \Rightarrow x^2+2 = A(x-1)(x+2)+Bx(x-1)+Cx(x+2)$$

and letting $x=0 \Rightarrow 2 = A(-1) \Rightarrow A=-1\,, \quad x=-2 \Rightarrow 4+2 = B(-2)(-3) \Rightarrow B=1$

and $x=1 \Rightarrow 1+2 = C(1)(3) \Rightarrow C=1$, so

$$\int \frac{x+2}{x^3+x^2-2x}\,dx = \int -\frac{1}{x}+\frac{1}{x+2}+\frac{1}{x-1}\,dx = -\ln x+\ln(x+2)+\ln(x-1)+C\,.$$

(iii) $\displaystyle\int_0^1 \frac{x}{\sqrt{1-x^2}}\,dx,$

SOLUTION

To solve the improper integral, replace the upper limit by k and after integrating, take limit $k \to 1$. Hence let $u = 1-x^2 \Rightarrow du = -2x\,dx$ and

$$\int \frac{x}{\sqrt{1-x^2}}\,dx = -\frac{1}{2}\int u^{-1/2}\,du = -\frac{1}{2}\frac{u^{1/2}}{1/2} = -\sqrt{u} = -\sqrt{1-x^2}\,.$$

Now apply limits of integration:

$$\int_0^k \frac{x}{\sqrt{1-x^2}}\,dx = -\left[\sqrt{1-x^2}\right]_0^k = -\sqrt{1-k^2}+\sqrt{1}$$

and finally take limit:

$$\int_0^1 \frac{x}{\sqrt{1-x^2}}\,dx = \lim_{k\to 1}\int_0^k \frac{x}{\sqrt{1-x^2}}\,dx = \lim_{k\to 1}\left(-\sqrt{1-k^2}+1\right) = 1\,.$$

(b) Integrate to show that [Seen similar/New]

$$\int \sqrt{1-x^2}\,dx = \frac{1}{2}x\sqrt{1-x^2}+\frac{1}{2}\sin^{-1}x+C\,.$$

SOLUTION

Need a substitution: answer as given suggests using

$x = \sin u \Rightarrow dx = \cos u\,du$ and $1-x^2 = 1-\sin^2 u = \cos^2 u \Rightarrow \sqrt{1-x^2} = \cos u$

so substituting into the integral we get

$$I = \int \sqrt{1-x^2}\,dx = \int \cos^u\,du = \frac{1}{2}\int \cos 2u+1\,du = \frac{1}{4}\sin 2u+\frac{1}{2}u+c = \frac{1}{2}\sin u\cos u+\frac{1}{2}u+c$$

and as $x = \sin u \Rightarrow u = \sin^{-1}x$ we have $I = \frac{1}{2}x\sqrt{1-x^2}+\frac{1}{2}\sin^{-1}x+c$, as required.

(c) Given the function $f(x) = \sqrt{1+x}$,

(i) Obtain the first four terms of the Maclaurin series of $f(x)$, and the Lagrange form of the remainder term R_3; [Seen similar]

SOLUTION

The Maclaurin series requires the function and four derivatives:

$f(x) = (1+x)^{1/2} \Rightarrow f(0) = 1\,,$

$f'(x) = \frac{1}{2}(1+x)^{-1/2} \Rightarrow f'(0) = \frac{1}{2}\,,$

$f''(x) = -\frac{1}{4}(1+x)^{-3/2} \Rightarrow f''(0) = -\frac{1}{4}\,,$

$$f^{(3)}(x) = \frac{3}{8}(1+x)^{-5/2} \Rightarrow f^{(3)}(0) = \frac{3}{8},$$
$$f^{(4)}(x) = -\frac{15}{16}(1+x)^{-7/2}\ .$$

The series is therefore

$$f(x) = f(0) + f'(0)x + \frac{f''(0)}{2}x^2 + \frac{f^{(3)}(0)}{6}x^3 + R_3 = 1 + \frac{1}{2}x - \frac{1}{8}x^2 + \frac{1}{16}x^3 + R_3$$

with remainder term

$$R_3(x) = -\frac{15}{16(1+c)^{7/2}}x^4\ ,$$

where $0 < |c| < |x|$.

(ii) If we use the series obtained in (i) to estimate $\sqrt{1.01}$, what is the maximal error incurred? Explain your reasoning. [New]

SOLUTION

To estimate $\sqrt{1.01} = \sqrt{1+0.01}$, we use $x = 0.01$ so $0 < c < 0.01$. Thus $(1+c)^{7/2}$ is smallest when $c = 0$, which gives the largest error, so the maximal error is

$$E_{\max} \ = \ |R_3| < \frac{15}{16}(0.01)^4 = \frac{15}{16} \times 10^{-8}\ .$$

4. (a) Find the radius of convergence of the following series: [Seen similar]

(i) $\displaystyle\sum_{n=1}^{\infty} \frac{\ln n}{n+2} x^n$

SOLUTION Again, the Ratio Test:

$$\left|\frac{a_{n+1}}{a_n}\right| = \left|\frac{(n+3)\,\ln(n+1)}{(n+2)\ln(n)}\right| |x|$$

Taking the limits:

$$\lim_{n\to\infty} \frac{n+3}{n+1} = \lim_{n\to\infty} \frac{1+\frac{3}{n}}{1+\frac{1}{n}} = 1$$

and for the other limit we use l'Hopital, case "∞/∞",

$$\lim_{n\to\infty} \frac{\ln(n+1)}{\ln n} = \lim_{n\to\infty} \frac{\frac{1}{n+1}}{\frac{1}{n}} = \lim_{n\to\infty} \frac{n}{n+1} = 1$$

which leaves

$$\lim_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right| = |x| < 1$$

for convergence, and the radius of convergence is $R = 1$.

(ii) $\displaystyle\sum_{n=1}^{\infty} \frac{x^{3n}}{n!}$

SOLUTION

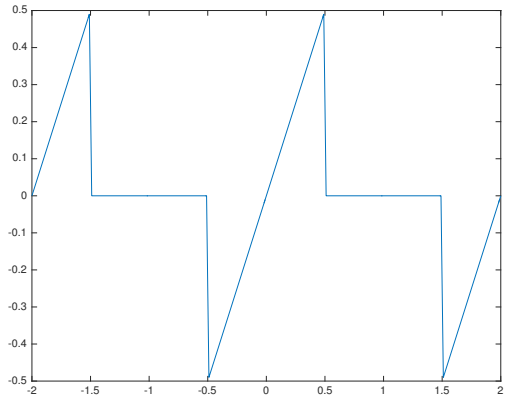
Ratio Test: $\displaystyle\lim_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right| = \lim_{n\to\infty} \left|\frac{n!\, x^{3(n+1)}}{(n+1)!\, x^{3n}}\right| = \lim_{n\to\infty} \frac{1}{n+1} |x^3| = 0 < 1$,
for all x, so the series converges for any x and has infinite radius of convergence.

(b) Given the function defined on the interval $[0,1)$ as

$$f(t) = \begin{cases} t & 0 \le t < 1/2 \\ 0 & 1/2 \le t < 1 \end{cases}$$

(i) If $g(t)$ is the *odd* extension of $f(t)$, sketch $g(t)$ for $-2 \le t \le 2$. [Seen similar]

SOLUTION



(ii) Show that the first eight terms of the Fourier Series for $g(t)$ are as follows [Seen similar]

$$g(t) = \frac{2}{\pi^2}\sin(\pi t) + \frac{1}{2\pi}\sin(2\pi t) - \frac{2}{(3\pi)^2}\sin(3\pi t) - \frac{1}{4\pi}\sin(4\pi t)$$

$$+\frac{2}{(5\pi)^2}\sin(5\pi t) + \frac{1}{6\pi}\sin(6\pi t) - \frac{2}{(7\pi)^2}\sin(7\pi t) - \frac{1}{8\pi}\sin(8\pi t) + \cdots$$

SOLUTION

The function $g(t)$ is odd, so a_n=0. The period is $T = 2$, so

$$b_n = \int_{-1}^{1} f(t)\ \sin(n\pi t)\ dt\ =\ 2\int_{0}^{1} f(t)\ \sin(n\pi t)\ dt\ =$$

using the half-range series; then

$$b_n = \int_0^{1/2} t\ \sin(n\pi t)\ dt\ + 2\int_{1/2}^{1} 0\ \sin(n\pi t)\ dt\ =\ 2\left[-t\frac{\cos(n\pi t)}{n\pi} + \frac{1}{n\pi}\int_0^{1/2}\cos(n\pi t)\ dt\right]$$

$$=\ 2\left[-t\frac{\cos(n\pi t)}{n\pi} + \frac{1}{(n\pi)^2}\sin(n\pi t)\right]_0^{1/2} =\ -\frac{1}{n\pi}\cos\left(\frac{n\pi}{2}\right) + 0 + \frac{2}{n^2\pi^2}\sin\left(\frac{n\pi}{2}\right) - 0$$ In the last expression we note that when n is odd, the cosine terms are zero, while the sine terms give ± 1. Hence

$$b_1 = \frac{2}{\pi^2}\ ,\ b_3 = -\frac{2}{3^2\pi^2}\ ,\ b_5 = \frac{2}{5^2\pi^2}\ \text{,and } b_7 = -\frac{2}{7^2\pi^2}\,.$$

Similarly, for even n the sine terms are zero, and the cosine terms give ± 1 so that

$$b_2 = \frac{1}{2\pi}\ ,\ b_4 = -\frac{1}{4\pi}\ ,\ b_6 = \frac{1}{6\pi}\text{, and }\ b_8 = -\frac{1}{8\pi}\ .$$

The Fourier series is

$$g(t) = \sum_{n=1}^{\infty} b_n \sin(n\pi t)$$

so the first eight terms are

$$g(t) = \frac{2}{\pi^2}\sin(\pi t) + \frac{1}{2\pi}\sin(2\pi t) - \frac{2}{(3\pi)^2}\sin(3\pi t) - \frac{1}{4\pi}\sin(4\pi t) + \frac{2}{(5\pi)^2}\sin(5\pi t) + \frac{1}{6\pi}\sin(6\pi t)$$

$$- \frac{2}{(7\pi)^2}\sin(7\pi t) - \frac{1}{8\pi}\sin(8\pi t) + \cdots$$

as required.

(iii) Choosing an appropriate value for t in the Fourier series for $g(t)$ above, obtain the value of the infinite sum [Seen similar/New]

$$\frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \frac{1}{7^2} + \frac{1}{9^2} \cdots$$

SOLUTION

To obtain the desired squares of odd integers in the denominators above, we clearly need only the odd b_n in the series, with even $b_n = 0$. Need to choose t so that

1) sine terms give correct $\pm$ pattern for odd n and

2) sine terms vanish for even $n = 2m$.

The last requirement suggests using $\sin(n\pi) = 0$, so in $\sin(2m\pi t)$, using $t = \frac{1}{2}$ works. In the series we now have

$$g(1/2) = \frac{2}{\pi^2}\sin(\pi/2) + 0 - \frac{2}{(3\pi)^2}\sin(3\pi/2) - 0 + \frac{2}{(5\pi)^2}\sin(5\pi/2) + 0 - \frac{2}{(7\pi)^2}\sin(7\pi/2) - 0 + \cdots$$

and we already used the idea of $\sin(n\pi/2) = \pm 1$ for odd n above, leaving

$$g(1/2) = \frac{2}{\pi^2}\left(\frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \frac{1}{7^2} + \frac{1}{9^2} + \cdots\right)$$

Finally, we recognize that $g(1/2)$ is at the discontinuity, where the Fourier series converges to the average of the two values 0 and $1/2$, hence

$$g(1/2) = \frac{1}{4} = \frac{2}{\pi^2}\left(\frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \frac{1}{7^2} + \frac{1}{9^2} + \cdots\right)$$

and the result for the series is $\pi^2/8$.